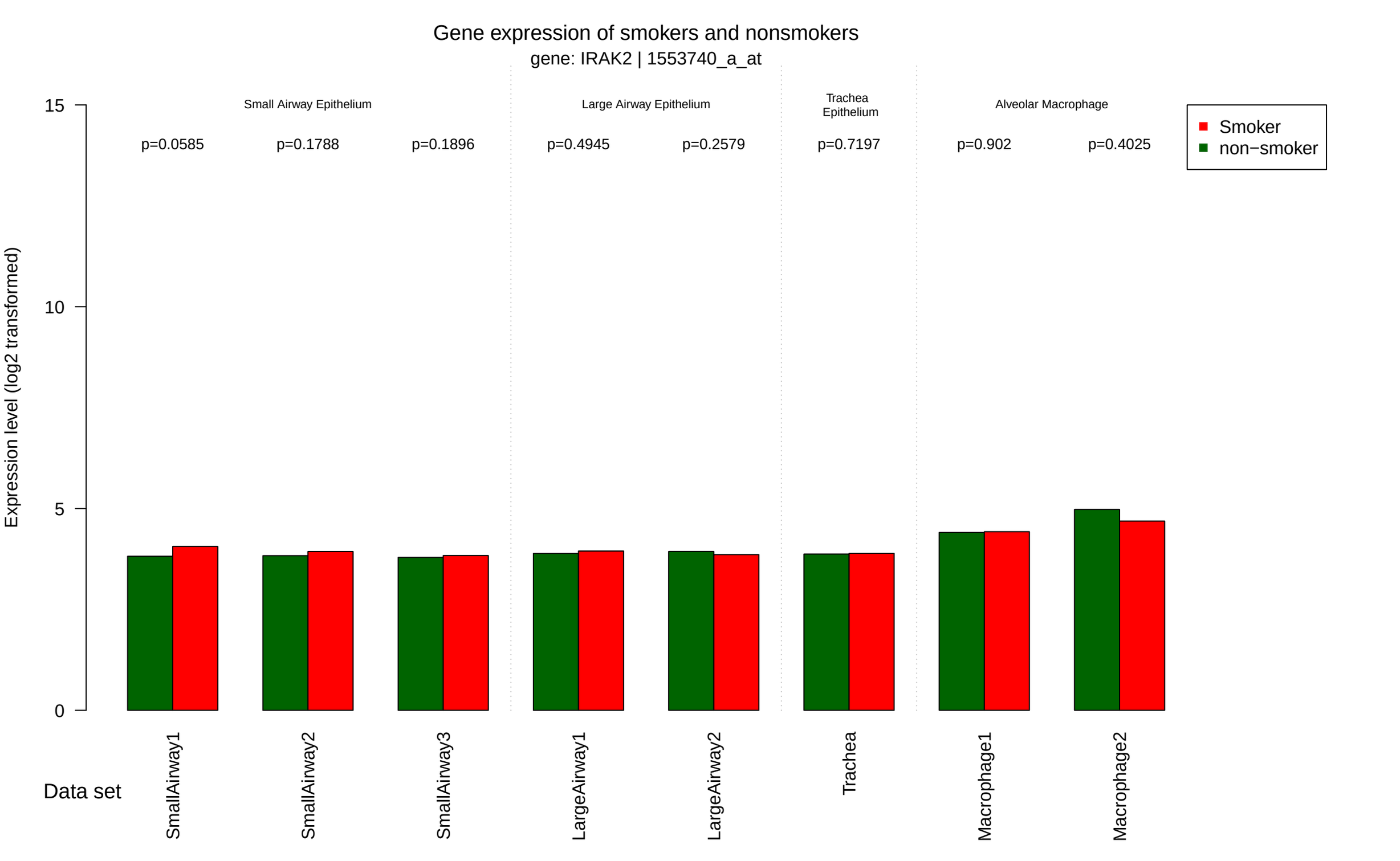
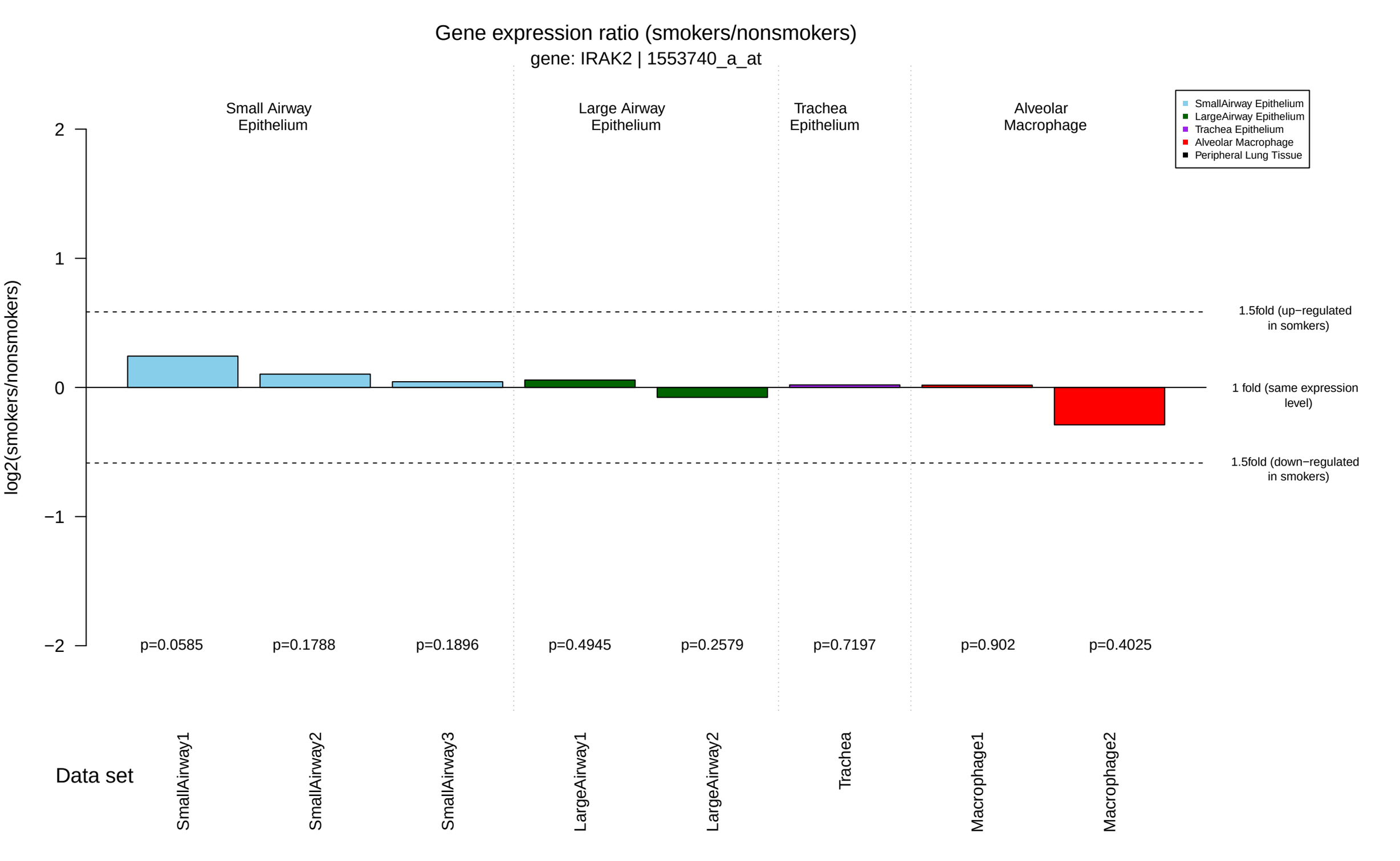
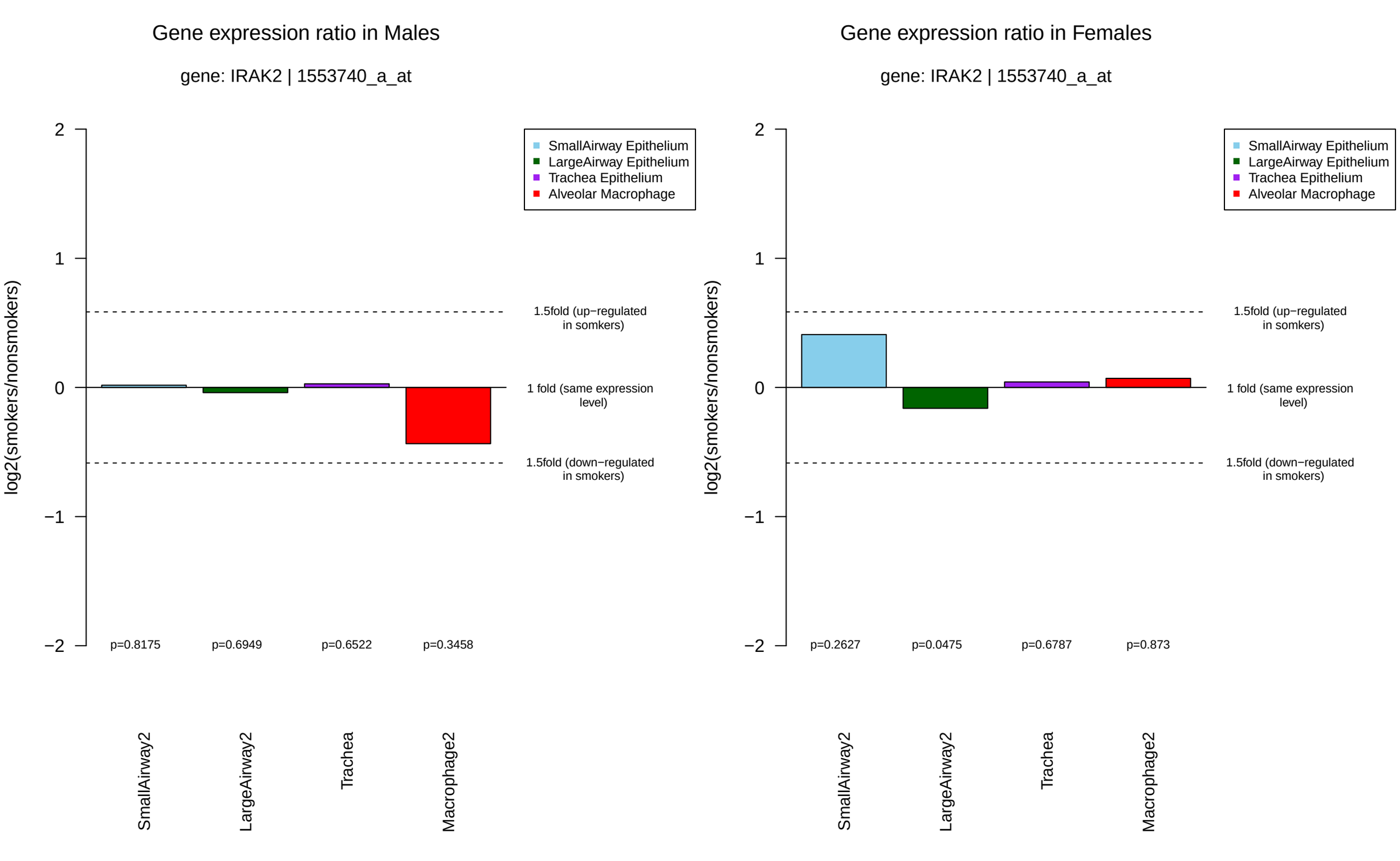
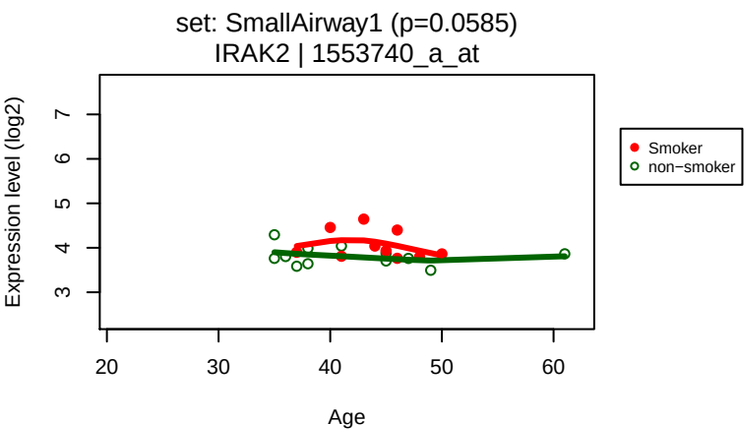
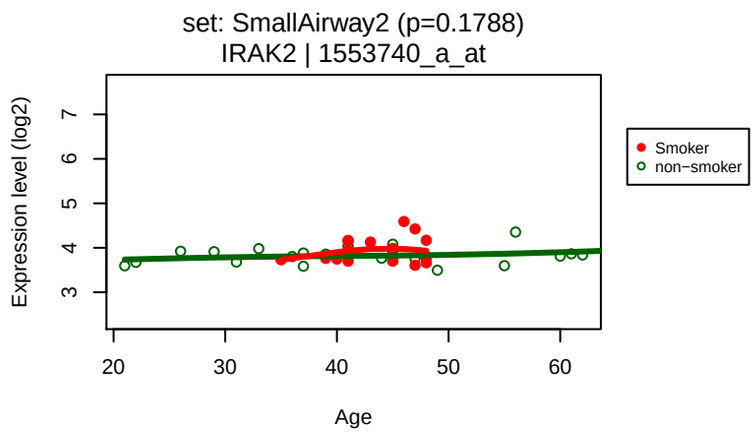
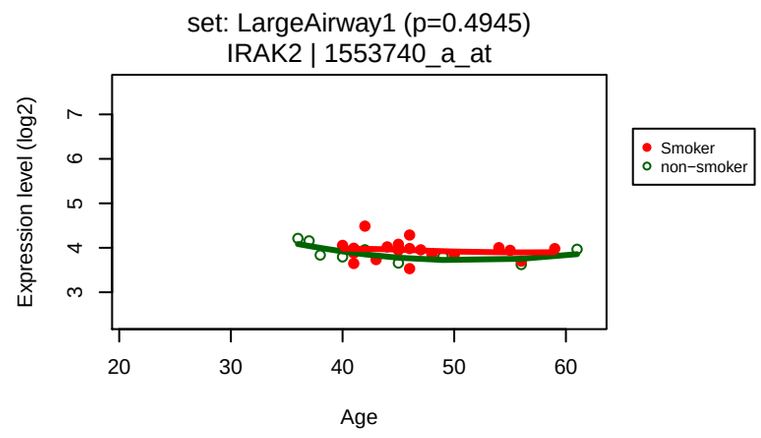

Gene expression of smokers and nonsmokers
gene: IRAK2 | 1553740_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0585
p=0.1788
p=0.1896
p=0.4945
p=0.2579
p=0.7197
p=0.902
p=0.4025
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

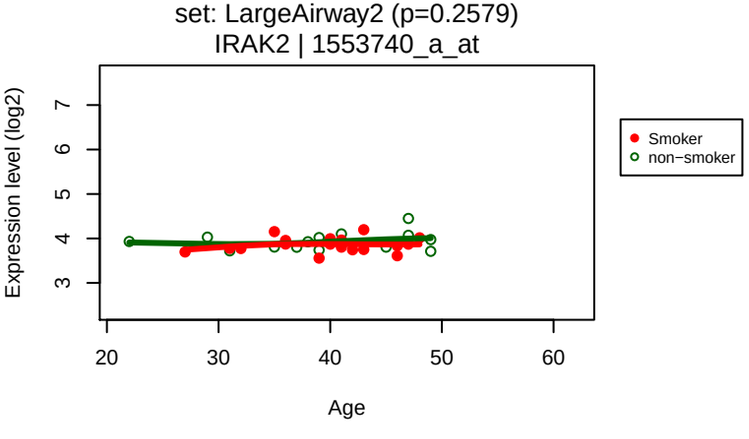

Gene expression ratio (smokers/nonsmokers)
gene: IRAK2 | 1553740_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0585
p=0.1788
p=0.1896
p=0.4945
p=0.2579
p=0.7197
p=0.902
p=0.4025
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

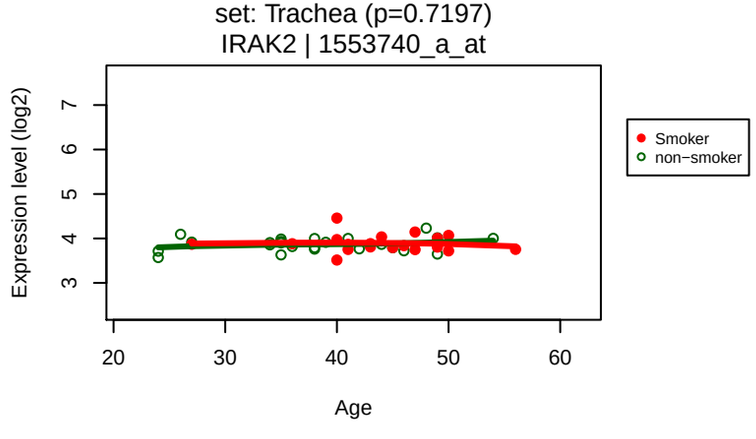

Gene expression ratio in Males
gene: IRAK2 | 1553740_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8175
p=0.6949
p=0.6522
p=0.3458
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRAK2 | 1553740_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2627
p=0.0475
p=0.6787
p=0.873
SmallAirway2
LargeAirway2
Trachea
Macrophage2

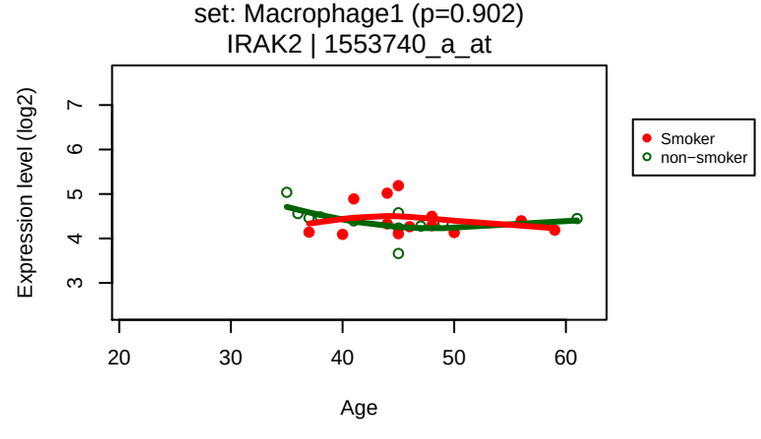

set: SmallAirway1 (p=0.0585)
IRAK2 | 1553740_a_at
Expression level (log2)
Age
Smoker
non−smoker

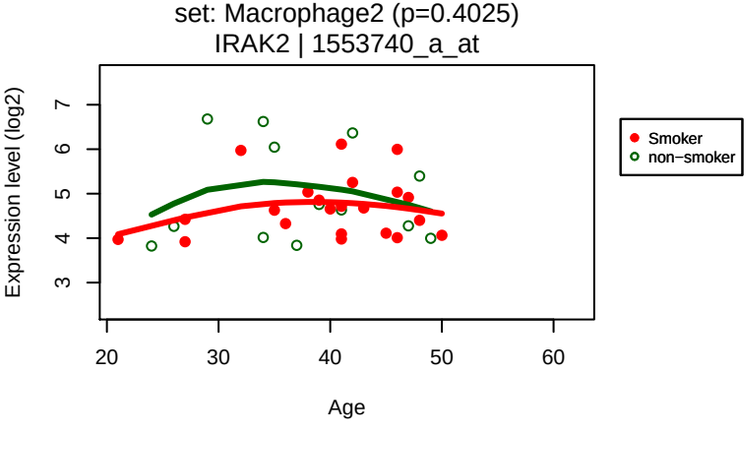

set: SmallAirway2 (p=0.1788)
IRAK2 | 1553740_a_at
Expression level (log2)
Age
Smoker
non−smoker

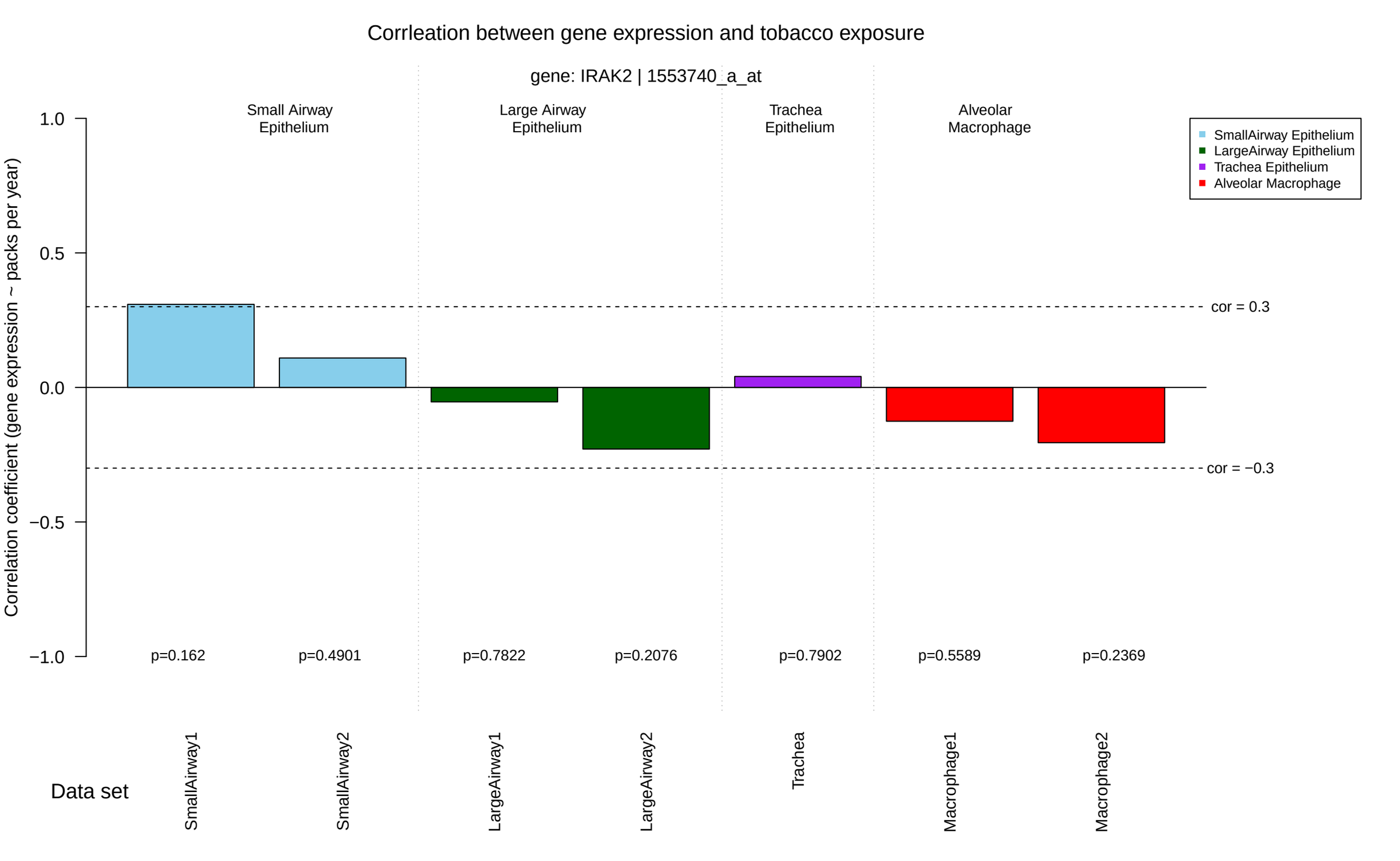

set: LargeAirway1 (p=0.4945)
IRAK2 | 1553740_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2579)
IRAK2 | 1553740_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7197)
IRAK2 | 1553740_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.902)
IRAK2 | 1553740_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4025)
IRAK2 | 1553740_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRAK2 | 1553740_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.162
p=0.4901
p=0.7822
p=0.2076
p=0.7902
p=0.5589
p=0.2369
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2